

Bopping and Swinging

What Goes On in the Jazz Brain?

When I listen to Joe Pass run lines on the guitar, I can tell him apart from other famous virtuosos like Django Reinhardt, Jim Hall, and Jodi Fisher. Often, however, when tuning into my favourite Indie Rock station, bands blend together as they borrow sounds and ideas from one another. Though these similarities define a genre, jazz brings in a level of personality and spontaneity other styles do not.

The relationship between theory and improvisation in jazz produces more than cool, experimental sounds; it actually stimulates various regions of the brain. In fact, playing jazz enhances connectivity between the left, right, and front hemispheres. The way an artist phrases notes on-the-fly depends a lot on what's going on in his or her head, making it easier for us to associate certain creative decisions with one artist and not another.

1 A CRASH COURSE IN THEORY

On a structural level, jazz differs from other music genres. Rhythmically, it swings: Musicians carry notes on the upbeat. In the 50s and 60s, this revolutionized Bebop. Although some artists now prefer smoother melodies, swing is still a prevalent jazz characteristic. Tone choices also contrast today's popular music by defying *keys*—a series of notes comprised of sharps (#), flats (b), and/or naturals (♮). To illustrate this point, let's compare a modern mainstream hit to an old-school standard.

1.1 COMPOSITION

Figure 1. *Afternoon in Paris*, John Lewis (1955):



Figure 1 shows the first eight bars of John Lewis' *Afternoon in Paris*, a staple jazz tune for student musicians. Since Lewis plays in the Key of C Major, the song should avoid sharps and flats. Bars two to six, however, include “accidentals”—notes outside of the defined key. In effect, the song moves from C Major to B $\flat$ Major in the third bar, then to A $\flat$ Major in the fifth. Although the song's written

one way, Lewis explores three different keys by challenging structure and expectation.

Figure 2. *Somebody I Used to Know*, Gotye (2011):



Figure 2 outlines Gotye's smash hit, *Somebody I Used to Know*. This tune is in the Key of F Major, meaning the song contains only one flat (B $\flat$). Unlike *Afternoon in Paris*, this song respects its key with a simple straight-forward melody, uninterrupted by accidentals.

1.2 IMPROVISATION

For harmonic accompaniment—what the rest of the band performs—Figures 1 and 2 are again different. The letters above the staff represent chords used in the song. This is known as a chord progression. *Somebody I Used to Know* uses strictly major (capital letters) and minor (lower-case letters) chords, both of which contain only three notes. *Afternoon in Paris*, on the other hand, uses “embellished” chords of four to six notes (i.e. E $\flat$ 9 and G13).

Embellished chords create alternate avenues for musicians to travel. This means a greater variety in melodic possibility. This also makes improvisation more

difficult. For *Somebody I Used to Know*, a soloist may pick any note within the key and write an acceptable solo. For *Afternoon in Paris*, the soloist must play to each chord rather than to the key, adding complexity and flavour. How a player chooses to approach these chords makes them identifiable as an artist.

2 CRANKING UP THE CREATIVITY

Inherently, improvisation is unpredictable. In the moment, artists think less about theory and more about sounds and feelings. So when considering chords and keys, these technicalities become mere skeletons to build from. On top of melody, other factors like loudness and timing come into play, which altogether strain the brain's frontal network. For instance, the amygdala—an area of the limbic system associated with emotion, behaviour, and long-term memory—activates when we listen to jazz musicians manipulate melody with rhythm and volume. When physically playing these variations, however, much more of the brain is at work.

2.1 TO THE LEFT; TO THE RIGHT

When improvising, jazz musicians use both hemispheres of the brain simultaneously. Independently, the left hemisphere is known for its analytical traits, such as data

processing, organization, and motor skills. Conversely, the right hemisphere includes all of the creativity. Since jazz improvisation is so technical, yet also free-form, both of these hemispheres work hard during a solo. Writing music over particularly tough rhythms puts both hemispheres into overtime.

3 WHEN INHIBITION STRAYS

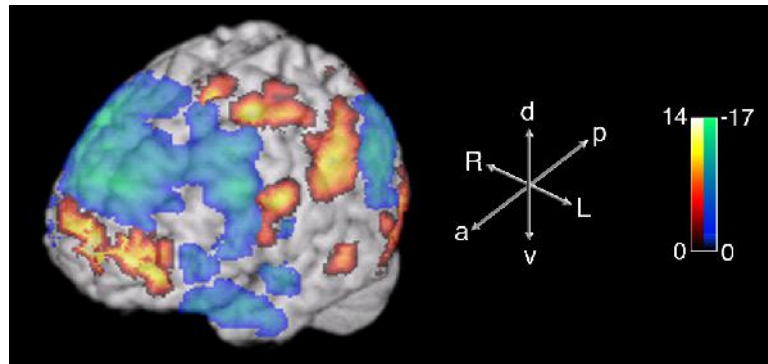
Scientists have studied how jazz affects the brain for years. Most recently, a group of Swedish researchers scanned 39 jazz pianists' brains while the musicians jammed on an MRI-friendly keyboard. The more experienced players revealed greater connectivity between three regions of the frontal lobe (recall the limbic system above) but less activity in areas that deal with organization and structure. The results of this study indicate a serious lack of inhibition during jazz improvisation.

Another study, published in PLoS ONE, examined a smaller group of artists and arrived at a similar conclusion. Using a special keyboard and headphones, scientists asked six trained jazz pianists to play inside of an MRI machine so that they could scan the brain during a jazz solo. This study followed four stages:

- (1) Pianists ascended (low to high) and descended (high to low) the notes of the C Major Scale;

- (2) Pianists improvised using only quarter notes (one beat) from the same scale;
- (3) Pianists performed an original blues tune against a backing track;
- (4) Pianists improvised over their original backing track.

Figure 3. Medial Prefrontal Cortex Activation, Limb and Braun (2011):



By the end of this procedure, the scientists noticed a trend. The dorsolateral cortex—the front region of the brain associated with memory, self-censoring, and planning—always slows down during a jazz solo. The very centre of the frontal lobe, however, known as the medial prefrontal cortex, spikes due to increased levels of self-expression.

While the process of composition would likely engage similar areas of the brain, jazz's heavy reliance on in-the-moment music demands a lot from musicians mentally. The act of studying jazz calls on the left hemisphere most, while playing recruits the right. Add fluctuations in rhythm and volume into the mix and the

frontal lobe activates, serving as a bridge to connect the left and right hemispheres. The result: A whole-brain workout.

References

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